

**DEFINITIVE PRODUCT SPECIFICATION & TECHNICAL
ARCHITECTURE**

Unite Attendance System

Complete end-to-end operational manual covering all 5 ecosystem applications, 13-stage lifecycle flow from signup to payroll, Prisma schema models, API matrix, security rules, and official subscription pricing.

All 5 Monorepo Apps

13 Operational Lifecycle Stages

All 8 Admin Sidebar Tabs

6 Settings Sub-Tabs

6 Super Admin Tabs

Official Pack Pricing (₹375 – ₹8,999/mo)

System Architecture & Monorepo Applications

Ecosystem
Engineering

apps/web

Website & Marketing

Public landing page, pricing pack selector, Razorpay subscription checkout, portal menu launcher, and product documentation.

apps/admin

Admin Dashboard

Organization portal (/dashboard, /attendance, /branches, /departments, /members, /qr-management, /reports, /settings) and Super Admin portal (/super-admin).

apps/app

Employee Mobile App

Mobile PWA for staff featuring geofenced GPS check-in/out, dynamic TOTP QR pass, history calendar, and profile badge.

apps/kiosk

Front-Desk Kiosk

Hardware tablet scanning terminal with sub-second camera QR verification, visual cards, audio chimes, and auto-reset.

apps/api

Backend API Gateway

NestJS REST API, WebSocket realtime gateway, Prisma ORM, PostgreSQL DB, JWT auth, and rules validation engine.

```
[ Employee Mobile App ] —(GPS + TOTP Pass)→ [ NestJS API Gateway ]  
←(Realtime WS)— [ Admin Dashboard ] | [ Front Desk Kiosk ] —(Sub-second  
Scan)→ [ Attendance Rules Engine ] → [ PostgreSQL & Prisma DB ]
```

Database Models & Entity Data Schemas

Prisma PostgreSQL
ERD

- ✓ **User:** id, email, name, avatarUrl, passwordHash, globalRole (SUPER_ADMIN | USER), isActive.
- ✓ **Organization:** id, name, slug, logo, plan (FREE | STARTER | PRO | ENTERPRISE), settings, isActive.
- ✓ **OrgMember:** id, userId, orgId, role (ORG_ADMIN | BRANCH_MANAGER | DEPT_HEAD | MEMBER), departmentId, branchId, employeeId, designation, phone.
- ✓ **Branch:** id, orgId, name, address, location (GPS Lat/Long JSON), isActive.
- ✓ **Department:** id, orgId, branchId, headId, name.

- ✓ **QRCode:** id, memberId, orgId, qrToken, type (MOBILE | ID_CARD), isActive, expiresAt.
- ✓ **AttendanceRecord:** id, memberId, orgId, type (CHECK_IN | CHECK_OUT), timestamp, method (QR_MOBILE | QR_IDCARD | QR_KIOSK | MANUAL), status (VALID | INVALID | FLAGGED), validationErrors, notes.
- ✓ **AttendanceRule:** id, orgId, workStart, workEnd, lateThresholdMin, halfDayThresholdMin, requireGps, gpsRadiusMeters, workingDays, preventDuplicateMin.
- ✓ **Subscription:** id, orgId, razorpaySubscriptionId, plan, status (ACTIVE | PAST_DUE | CANCELLED | TRIALING).

End-to-End Operational Lifecycle

(13 Stages)

Signup to Payroll
Export

Stage 1: Website Landing Page & Pack Selection

apps/web

Public Website & Pack Selection

The client visits the public web application, explores ecosystem features, and selects an active employee strength pack tier.

- ✓ Choose Pack Tier: Starter 10 (₹375/mo), Starter 25 (₹749/mo), Business 50 (₹1,125/mo), Pro Growth 100 (₹1,875/mo), Scale 200 (₹2,999/mo), Corporate 500 (₹5,249/mo), Enterprise 1000 (₹8,999/mo).
- ✓ Launch Discount: Flat 25% OFF billed monthly.
- ✓ 14-Day Free Trial: Full feature access with no credit card required.

uniteattendance.com/#pricing

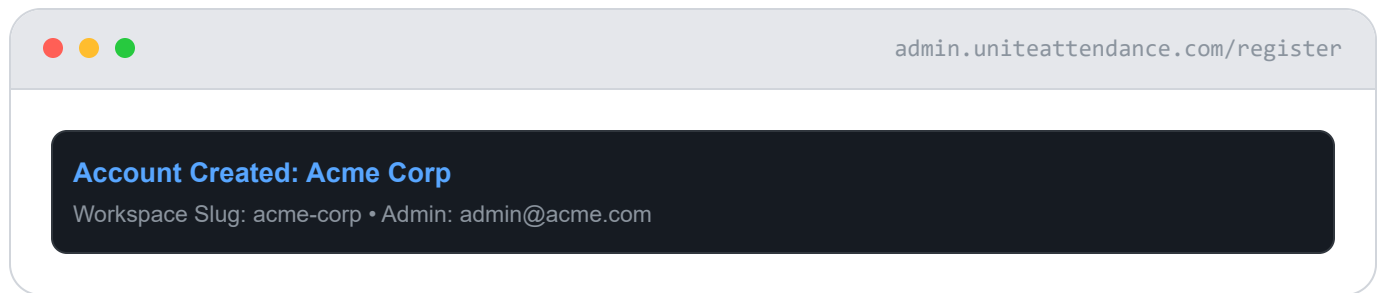
Pro Growth 100 Pack ★ (Popular)

51 – 100 Employees • Launch Price: ₹1,875/mo (₹18.75/emp)

Account Signup & Organization Slug Creation

Organization Admin registers the corporate account. The system automatically provisions a tenant workspace with a unique slug.

- ✓ Input Company Name, Admin Name, Corporate Email, Password.
- ✓ Automated Organization Slug allocation (e.g., acme-corp).
- ✓ Razorpay checkout integration for subscription activation.



Corporate Identity & Legal Entity Registration

Admin logs in and completes corporate profile details in the Settings PROFILE sub-tab.

- ✓ Organization Display Name & Legal Registered Entity Name.
- ✓ Industry Sector & Company Size Selection.
- ✓ Corporate Email, Phone, Website URL, GSTIN / Tax ID, HQ Address.
- ✓ Timezone (Asia/Kolkata IST) & Currency (INR ₹) defaults.



admin.uniteattendance.com/settings?tab=PROFILE

Legal Entity: Unite Technologies Pvt. Ltd.
GSTIN: 27AAACU1234M1Z5 | Timezone: Asia/Kolkata

Shift Hours, Grace Period & Geofence Configuration

Defines corporate work hours, late arrival grace thresholds, working days, and spatial GPS geofencing parameters.

- ✓ Shift Start (09:00 AM) & Shift End (06:00 PM).
- ✓ Late Grace Threshold (15 mins) & Half-Day Criteria (240 mins).
- ✓ Working Days Selector (Mon, Tue, Wed, Thu, Fri).
- ✓ Mandatory GPS Geofencing toggle & Allowed Radius (e.g. 200m).



admin.uniteattendance.com/settings?tab=RULES

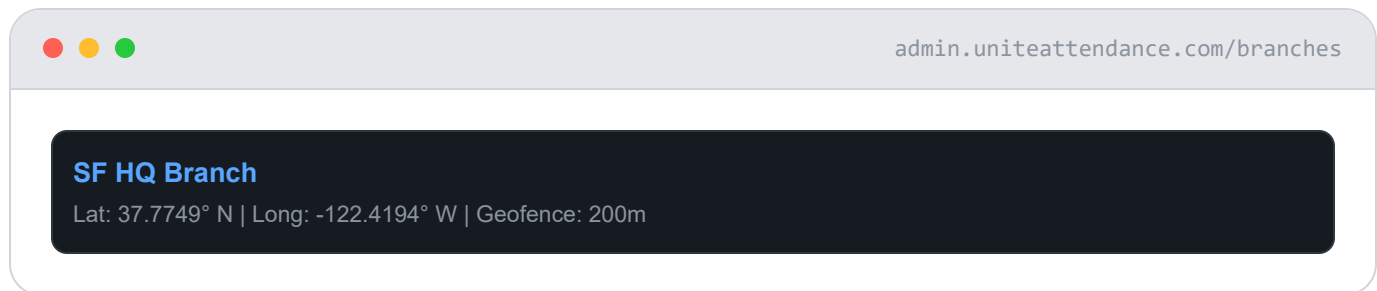
Shift: 09:00 - 18:00 | Grace: 15m

Require GPS: True | Radius: 200 Meters

Multi-Branch Mapping & Geofence Coordinates

Admins create physical office locations and configure latitude/longitude coordinates for mobile GPS verification.

- ✓ Branch Name, Physical Address, Active status toggle.
- ✓ GPS Latitude & Longitude coordinates.
- ✓ Branch Geofence radius in meters per office.



Department Structuring & Management Mapping

Creates organizational departments and maps Department Heads to oversee team attendance.

- ✓ Create Engineering, Operations, Sales, HR departments.
- ✓ Department Head Assignment: Grant managers access to team reports.
- ✓ Member Count tracking per department.



admin.uniteattendance.com/departments

Engineering Dept (42 Members)

Head: Sarah Miller

Employee Directory & Access Role Mapping

Admins populate employee directory via modal or CSV bulk import, assigning roles and branch/department mappings.

- ✓ Add Single Member or Bulk Import via CSV sheet.
- ✓ Assign Roles: ORG_ADMIN, BRANCH_MANAGER, DEPT_HEAD, MEMBER.
- ✓ Designation, Employee ID, Phone, Branch, and Dept mapping.



admin.uniteattendance.com/members

Alex Johnson (EMP-88492)

Senior Engineer • Engineering Dept • Role: MEMBER

Dynamic QR Pass & Physical ID Sheet Export

Generates encrypted QR tokens for staff, available dynamically in mobile app or as printable ID card sheets.

- ✓ Mobile Dynamic Pass: Time-rotating TOTP QR code preventing screenshot sharing.
- ✓ Printable ID Cards: Batch export ID card sheets for lanyard badges.
- ✓ Admin Reset Password: One-click temp password generation for staff.
- ✓ Bulk QR Token Regeneration: Instant token refresh across active staff.



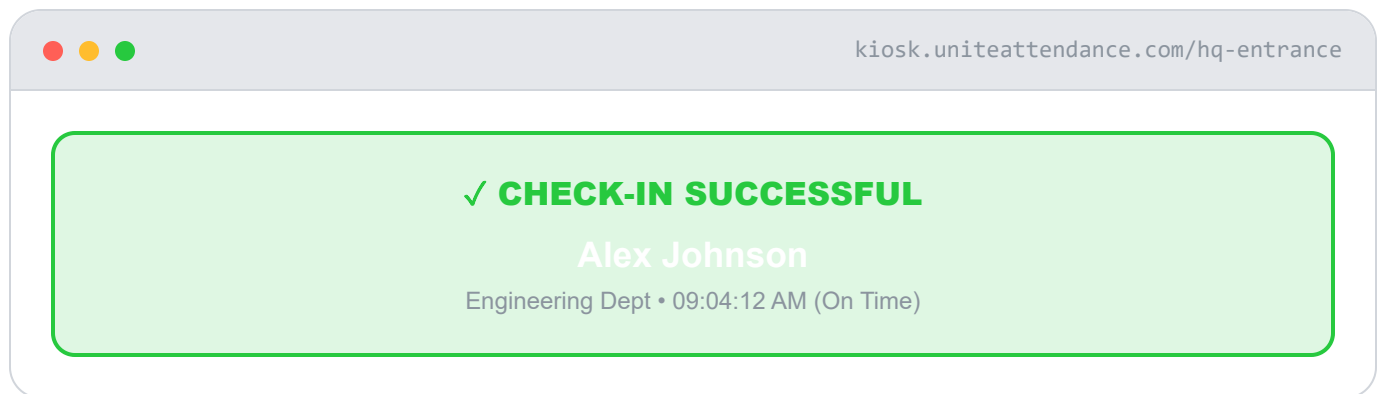
admin.uniteattendance.com/qr-management

[Print Batch ID Card Sheet \(PDF\)](#)

Entrance Camera Scanner Terminal

Mounted tablets or front-desk computers run the Kiosk application, continuously scanning incoming employee QR codes.

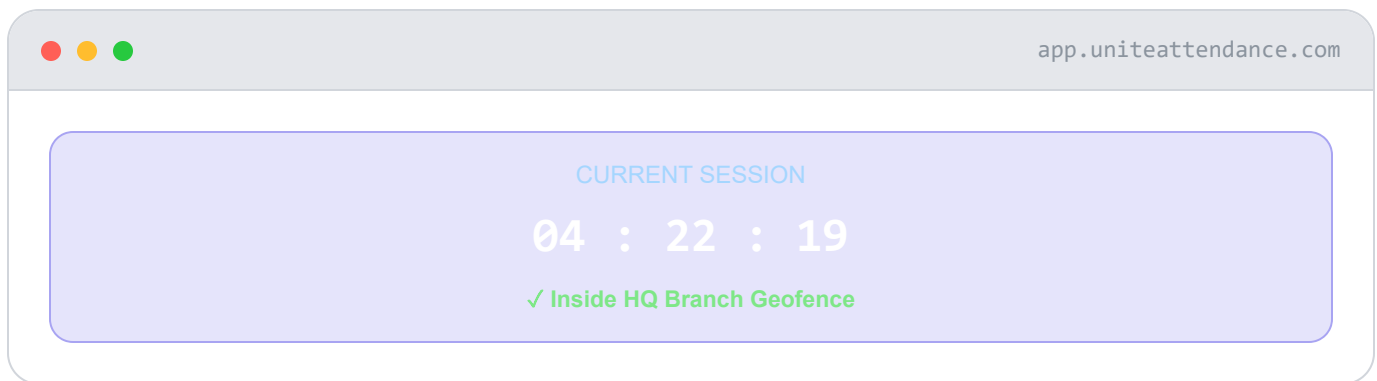
- ✓ Sub-second scan processing (<300ms verification latency).
- ✓ Instant visual card: Green banner with photo & name for success, Red for invalid/expired.
- ✓ Audio chime confirmation for hands-free visual acknowledgment.
- ✓ Automatic 3-second auto-reset for seamless queue management.



Staff Mobile Self-Service Check-In & Digital Badge

Employees use the mobile app for self-service check-in with GPS verification, displaying dynamic QR pass, and reviewing history.

- ✓ **Home Tab (/):** Active session timer, geofenced GPS check-in/out button, distance to branch indicator.
- ✓ **My QR Pass Tab (/qr):** Dynamic TOTP QR code pass with live countdown timer.
- ✓ **History Tab (/history):** Personal monthly attendance log, status badges (Valid, Late, Early Exit).
- ✓ **Profile Tab (/profile):** Digital ID badge & assigned department/branch details.



WebSocket Stream & Flagged Log Resolution

Admins monitor live WebSocket check-in feeds, inspect status flags, and apply manual attendance corrections.

- ✓ WebSocket Live Feed displaying live check-in counters & status metrics.
- ✓ Status Flags: VALID (Green), INVALID (Red), FLAGGED (Yellow).
- ✓ Manual Override: Apply check-in corrections with mandatory admin audit notes.

admin.uniteattendance.com/attendance			
MEMBER	TIME	METHOD	STATUS
Alex Johnson	09:04 AM	QR_KIOSK	VALID
Sarah Miller	09:32 AM	QR_MOBILE	FLAGGED (Late)

Monthly Register Matrix & Payroll Export

Admins generate department attendance summaries, monthly registers, and export payroll-ready CSV & PDF files.

- ✓ Daily Summary & Monthly Register matrix views.
- ✓ Department punctuality breakdown & attendance rate %.
- ✓ 1-Click Export to CSV spreadsheet & printable PDF register.



admin.uniteattendance.com/reports

Monthly Register Ready (Aug 2026)

150 Active Employees • 3,120 Scans Processed

Platform Tenant Control & Aura AI Anomaly Engine

Platform owners manage global organization tenants, subscription plans, AI anomaly engines, and platform audit trails.

- ✓ **Platform Overview (/super-admin/dashboard):** Global tenant count, subscription status, revenue.
- ✓ **Organizations (/super-admin/organizations):** Global tenant list, direct plan upgrades/downgrades.
- ✓ **Aura AI Control (/super-admin/aura):** AI engine configuration & anomaly detection insights.
- ✓ **Global Audit Logs (/super-admin/audit-logs):** Platform-wide security audit trail.



admin.uniteattendance.com/super-admin

Super Admin Portal Mode

Managing 42 Active Organization Tenants

Official Subscription Packs & Pricing Structure

Flat 25%
Launch Offer

Unite Attendance offers transparent, pack-based pricing tailored to your active employee count. All plans include a **14-Day Free Trial** with no credit card required.

PRICING PACK NAME	EMPLOYEE RANGE	REGULAR PRICE	LAUNCH PRICE (25% OFF)	EFFECTIVE RATE
Starter 10 Pack	Up to 10 Employees	₹499/mo	₹375 / month	₹37.50 / employee
Starter 25 Pack	11 – 25 Employees	₹999/mo	₹749 / month	₹30.00 / employee
Business 50 Pack	26 – 50 Employees	₹1,499/mo	₹1,125 / month	₹22.50 / employee
Pro Growth 100 Pack ★ (POPULAR)	51 – 100 Employees	₹2,499/mo	₹1,875 / month	₹18.75 / employee
Scale 200 Pack	101 – 200 Employees	₹3,999/mo	₹2,999 / month	₹15.00 / employee
Corporate 500 Pack	201 – 500 Employees	₹6,999/mo	₹5,249 / month	₹10.50 / employee
Enterprise 1000 Pack	501 – 1,000 Employees	₹11,999/mo	₹8,999 / month	₹9.00 / employee
Large Enterprise	1,001+ Employees	Custom Quote / Sales Contact (Dedicated SLA & Multi-city deployment)		

✓ Multi-Branch & Dept Support

✓ Role-Based Manager Access

✓ 1-Click PDF & CSV Exports

✓ Mobile App & Kiosk Web PWA

✓ Dynamic TOTP Pass Scanning

✓ Shift & Grace Period Rules

✓ 100% Cloud Access (No PC App)

✓ Audit Log & Security Trail

© 2026 Unite Attendance Platform. All rights reserved.

Official Comprehensive Client Product Manual & Lifecycle Specification • Document Version 6.0